

# **Calibration, Scoring Rules, and Conformal Prediction: Complete Solutions**

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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# 1 Part I – Calibration and Scoring Rules

## A: Conceptual Questions

### A1. Proper vs. Improper Scoring Rules

#### Solution

A scoring rule  $S(p_\theta, (x, y))$  is **proper** if the expected score is optimised when the predictive distribution matches the true conditional distribution  $p^*(y | x)$ :

$$\mathbb{E}_{(x,y) \sim p^*} [S(p_\theta, (x, y))] \leq \mathbb{E}_{(x,y) \sim p^*} [S(p^*, (x, y))],$$

with equality if and only if  $p_\theta = p^*$  (strictly proper). A rational forecaster minimising the expected loss therefore has no incentive to misreport beliefs.

**Why NLL is proper.** For a binary event with true probability  $q$  and reported probability  $p$ , the expected NLL is  $L(p) = -[q \log p + (1 - q) \log(1 - p)]$ . Taking  $L'(p) = 0$  gives  $p = q$  (proved in full in B1 below). Because the minimum is attained exactly at the truth, NLL incentivises honest reporting.

**Why the rank reward is not proper.** If we reward a forecaster for simply placing the true label first—regardless of the reported probability—the forecaster is incentivised to assign probability  $1 - \epsilon$  to a single class and  $\epsilon/(C - 1)$  to all others. The rule never penalises overconfidence, so it does not force  $p_\theta = p^*$  and hence is not proper.

### A2. NLL vs. Brier Score: Sensitivity Comparison

#### Solution

**NLL** penalises via  $-\log p$ , which diverges to  $+\infty$  as  $p \rightarrow 0$ . Even a single catastrophically confident wrong prediction can dominate the average loss. NLL is *tail-sensitive*.

**Brier score** penalises via squared error  $(\hat{p}_c - \mathbf{1}[y = c])^2$ , which is bounded in  $[0, 1]$ . Extreme probabilities are penalised more than moderate ones, but the penalty remains finite. The metric is outlier-robust.

**Practical guidance:** Prefer **NLL** for medical diagnosis, safety-critical decisions, and information-theoretic model comparison, where overconfident wrong predictions must be heavily penalised. Prefer **Brier score** for weather forecasting and settings where a bounded, interpretable, outlier-robust metric is needed.

### A3. Calibration vs. Sharpness

#### Solution

**Calibration.** A model is calibrated if predicted probabilities match empirical frequencies:  $p^*(y = 1 | \hat{p} = c) \approx c$  for all  $c \in [0, 1]$ . A perfectly calibrated reliability diagram lies on the diagonal.

**Sharpness.** A model is sharp if its forecasts are concentrated near 0 or 1 (decisive predictions), rather than being spread near the base rate.

**Calibrated but poor:** A model that always predicts the marginal base rate  $\bar{y}$  is perfectly calibrated (the constant prediction matches the population frequency) yet carries no

discriminative information.

**Sharp but miscalibrated:** A deep neural network trained with cross-entropy often outputs very high softmax probabilities (sharp) while those probabilities substantially exceed the empirical accuracy (overconfident reliability diagram lies below the diagonal; Guo et al., ICML 2017).

The ideal forecaster is both calibrated *and* sharp.

#### A4. ECE: Definition and Limitations

##### Solution

(a) With  $B$  bins, let  $\mathcal{B}_b$  be the set of test indices whose top-1 confidence falls in bin  $b$ :

$$\text{acc}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} \mathbf{1}(\hat{y}_i = y_i), \quad \text{conf}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} \hat{p}_i.$$

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{N} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|.$$

This is a weighted average of per-bin calibration gaps, weighted by bin population.

(b) **Limitations:**

- (i) **Binning sensitivity.** ECE changes with the number of bins and whether bins have equal width or equal frequency; there is no canonical definition.
- (ii) **Top-1 only.** ECE evaluates only the top predicted class, ignoring miscalibration of the remaining class probabilities.

(c) **Alternatives:** Classwise/marginal calibration error (checks all class probabilities), adaptive equal-frequency binning, differentiable calibration losses, and full proper scoring rules (NLL, Brier) that require no binning.

#### A5. Why Modern Deep Networks Are Miscalibrated

##### Solution

**1. Network depth and capacity.** Large networks can fit training data with very high softmax probabilities. The final softmax magnifies small logit differences, pushing predictions toward 0 or 1. Because NLL-based training only heavily penalises wrong confident predictions (which become rare as the model improves), the model has little incentive to avoid global overconfidence.

**2. Batch Normalization and weight decay.** BatchNorm normalises intermediate activations, which can allow logits to grow large; weight decay reduces weights but does not directly control the logit scale entering the softmax. Together they can produce sharply peaked softmax outputs that are not well-aligned with true posterior probabilities.

**Intuitive summary.** Training minimises finite-sample NLL; the optimum of this objective is achieved by confident correct predictions on training data. The model has no incentive to spread probability mass to other classes on in-distribution inputs, leading to overconfident

test-set predictions.

## A6. Temperature Scaling: Mechanism and Properties

### Solution

(a)

$$\hat{\mathbf{q}}(\mathbf{z}; T) = \text{softmax}\left(\frac{\mathbf{z}}{T}\right), \quad \hat{q}_c = \frac{\exp(z_c/T)}{\sum_{c'} \exp(z_{c'}/T)}.$$

(b)  $T > 1$ : divides logits by a large value, reducing their magnitude; the softmax becomes softer (more uniform) – *reduces overconfidence*.  $T < 1$ : amplifies logits, sharpening the softmax – *increases confidence*.

(c) The argmax of  $\mathbf{z}/T$  equals the argmax of  $\mathbf{z}$  because dividing all components by the same positive scalar  $T$  preserves their relative ordering. Temperature scaling therefore leaves the top-1 predicted class unchanged.

(d) The single free parameter is  $T > 0$  (often parameterised as  $\log T$  for unconstrained optimisation). It is fitted by minimising NLL on a held-out *calibration* split, separate from both training and test data.

## B: Mathematical Derivations

### B1. Strict Propriety of NLL

#### Solution

(a) Differentiating  $L(p) = -[q \log p + (1 - q) \log(1 - p)]$ :

$$L'(p) = -\frac{q}{p} + \frac{1 - q}{1 - p} = \frac{-q(1 - p) + p(1 - q)}{p(1 - p)} = \frac{p - q}{p(1 - p)}.$$

Since  $p(1 - p) > 0$  for  $p \in (0, 1)$ :

$$\boxed{L'(p) = 0 \iff p = q.}$$

(b)

$$L''(p) = \frac{q}{p^2} + \frac{1 - q}{(1 - p)^2}.$$

Both terms are strictly positive for  $p \in (0, 1)$  and  $q \in (0, 1)$ , so  $\boxed{L''(p) > 0}$  for all valid  $p$ .

(c) **Interpretation.** The unique global minimum is at  $p = q$ . A rational forecaster who minimises expected NLL has no incentive to report anything other than the true probability – the scoring rule is *strictly proper* and elicits honest beliefs.

## B2. Strict Propriety of the Brier Score

### Solution

(a) Expanding  $B(p)$ :

$$\begin{aligned} B(p) &= q(p-1)^2 + (1-q)p^2 \\ &= q(p^2 - 2p + 1) + (1-q)p^2 \\ &= p^2[q + (1-q)] - 2qp + q \\ &= p^2 - 2qp + q. \end{aligned}$$

This is an upward-opening quadratic in  $p$ :

$$B'(p) = 2p - 2q = 0 \implies \boxed{p^* = q}.$$

The minimum is attained at the true probability, confirming strict propriety.

(b) Minimum value:

$$B(q) = q^2 - 2q^2 + q = q - q^2 = \boxed{q(1-q)}.$$

This equals the *variance of a Bernoulli*( $q$ ) random variable: the irreducible uncertainty that remains even when the forecaster reports the truth. It is maximised at  $q = 0.5$  and equals zero at certainty ( $q \in \{0, 1\}$ ).

## B3. ECE Decomposition

### Solution

(a) The ECE formula is already in the form of a weighted mean:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |a_b - c_b| = \mathbb{E}_b[|a_b - c_b|],$$

where  $n_b/N$  is the weight of bin  $b$  and  $|a_b - c_b|$  is the per-bin calibration gap.

(b) With  $B = 1$  all samples fall in one bin, so:

$$\text{ECE} = |\bar{a} - \bar{c}|,$$

where  $\bar{a}$  is overall accuracy and  $\bar{c}$  is mean predicted confidence. This measures only the *global* over/under-confidence, missing all within-range miscalibration.

(c) Perfect accuracy means  $a_b = 1$  for all  $b$ . If  $c_b < 1$  for some bin:  $|a_b - c_b| = 1 - c_b > 0$ , so ECE is positive. A model can be perfectly accurate yet poorly calibrated (underconfident): it always predicts the correct class but assigns low probability to it.

#### B4. Temperature Scaling as a 1-D Optimisation

##### Solution

(a) Substituting  $s = 1/T$  (so  $z_{ic}/T = s z_{ic}$ ):

$$\hat{q}_c(\mathbf{z}; s) = \frac{\exp(s z_c)}{\sum_{c'} \exp(s z_{c'})}.$$

The loss becomes

$$\mathcal{L}(s) = -\frac{1}{M} \sum_{i=1}^M \left( s z_{i,y_i} - \log \sum_{c'} \exp(s z_{ic'}) \right).$$

(b) Differentiating term by term:

$$\frac{\partial \mathcal{L}}{\partial s} = -\frac{1}{M} \sum_{i=1}^M \left( z_{i,y_i} - \underbrace{\sum_{c'} \hat{q}_{c'}(\mathbf{z}_i; s) z_{ic'}}_{\mathbb{E}_{\hat{q}_i}[z_c]} \right).$$

**Optimality condition:**  $\partial \mathcal{L} / \partial s = 0$  when the average true-class logit equals the average expected logit under the calibrated distribution.

(c) The function  $f(s) = \log \sum_{c'} \exp(s z_{c'})$  (log-sum-exp of linear functions of  $s$ ) is convex in  $s$ . The loss  $\mathcal{L}$  is a non-negative linear combination of such terms, hence **convex in**  $s = 1/T$ . Note that convexity in  $s$  does not automatically imply convexity in  $\tau = \log T$  (since  $s = e^{-\tau}$  is not an affine function of  $\tau$ ). However, the mapping  $s \mapsto \tau = -\log s$  is monotone, so the unique global minimum in  $s$  corresponds to a unique global minimum in  $\tau$ . Parameterising as  $\tau = \log T$  (unconstrained) guarantees  $T > 0$  and permits standard gradient-based optimisation to find the unique minimum.

#### B5. NLL as Entropy plus KL Divergence

##### Solution

(a) Definitions.

| Quantity         | Formula and $\theta$ -dependence                                                                                                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Entropy of $p^*$ | $H(p^*) = - \sum_y p^*(y) \log p^*(y)$<br><b>Does not depend on <math>\theta</math>.</b>                                                                                                                    |
| KL divergence    | $\text{KL}(p^* \  p_\theta) = \sum_y p^*(y) \log \frac{p^*(y)}{p_\theta(y)}$<br><b>Depends on <math>\theta</math>.</b>                                                                                      |
| Cross-entropy    | $H(p^*, p_\theta) = - \sum_y p^*(y) \log p_\theta(y)$<br><b>Depends on <math>\theta</math>.</b>                                                                                                             |
| Expected NLL     | $\mathcal{L}(\theta) = \mathbb{E}_{y \sim p^*} [-\log p_\theta(y)] = - \sum_y p^*(y) \log p_\theta(y)$<br><b>Depends on <math>\theta</math>. Note: <math>\mathcal{L}(\theta) = H(p^*, p_\theta)</math>.</b> |

**(b) Four-step derivation.**

**Step 1.** Write the expected NLL explicitly:

$$\mathcal{L}(\theta) = - \sum_y p^*(y) \log p_\theta(y).$$

**Step 2.** Add and subtract  $\log p^*(y)$  inside the sum (a legal  $\pm$  zero trick):

$$\mathcal{L}(\theta) = - \sum_y p^*(y) \left[ \log p^*(y) - \log p^*(y) + \log p_\theta(y) \cdot (-1) \right] = - \sum_y p^*(y) \log p^*(y) + \sum_y p^*(y) \log \frac{p^*(y)}{p_\theta(y)}.$$

Wait — let us be careful with signs. Splitting the single sum:

$$\begin{aligned} \mathcal{L}(\theta) &= - \sum_y p^*(y) \log p_\theta(y) \\ &= - \sum_y p^*(y) \log p^*(y) + \sum_y p^*(y) \log p^*(y) - \sum_y p^*(y) \log p_\theta(y) \\ &\quad \text{(add } \pm \sum p^* \log p^*) \\ &= - \sum_y p^*(y) \log p^*(y) + \sum_y p^*(y) \log \frac{p^*(y)}{p_\theta(y)}. \quad \text{(combine last two sums)} \end{aligned}$$

**Step 3.** Recognise each piece:

$$- \sum_y p^*(y) \log p^*(y) = H(p^*), \quad \sum_y p^*(y) \log \frac{p^*(y)}{p_\theta(y)} = \text{KL}(p^* \| p_\theta).$$

**Step 4.** Conclude:

$$\boxed{\mathcal{L}(\theta) = H(p^*) + \text{KL}(p^* \| p_\theta)}.$$

**(c) Gibbs' inequality and consequences.**

**Gibbs' inequality:** For any two probability distributions  $p$  and  $q$ ,

$$\text{KL}(p\|q) = \sum_y p(y) \log \frac{p(y)}{q(y)} \geq 0, \quad \text{with equality iff } q = p \text{ (pointwise).}$$

*Proof sketch:* Apply Jensen's inequality to the strictly concave function  $f(t) = -\log t$ :  $\mathbb{E}[-\log(q/p)] \geq -\log(\mathbb{E}[q/p]) = -\log 1 = 0$ .

(i) From the decomposition and Gibbs:

$$\mathcal{L}(\theta) = H(p^*) + \underbrace{\text{KL}(p^*\|p_\theta)}_{\geq 0} \geq H(p^*),$$

with equality iff  $\text{KL}(p^*\|p_\theta) = 0$  iff  $p_\theta = p^*$ . The minimum achievable NLL is the entropy of the true distribution: an irreducible floor that no model can beat.

(ii) Since  $H(p^*)$  is a constant with respect to  $\theta$ :

$$\arg \min_{\theta} \mathcal{L}(\theta) = \arg \min_{\theta} [H(p^*) + \text{KL}(p^*\|p_\theta)] = \arg \min_{\theta} \text{KL}(p^*\|p_\theta).$$

**Minimising expected NLL is exactly minimising the KL divergence from  $p^*$  to  $p_\theta$ .** This is the information-theoretic content of maximum likelihood estimation.

**(d) Link to maximum likelihood.**

Minimising the average NLL  $-\frac{1}{N} \sum_{i=1}^N \log p_\theta(y_i)$  is equivalent to maximising  $\frac{1}{N} \sum_i \log p_\theta(y_i)$ , which is the log-likelihood divided by  $N$ . Monotone transformations ( $\times N$ , then  $\exp$ ) preserve the argmax, so this is equivalent to maximising  $\prod_{i=1}^N p_\theta(y_i)$  – the likelihood under i.i.d. data.

**(e) Strict propriety in one sentence.**

The expected log score satisfies  $\mathbb{E}_{p^*}[\log p_\theta] = -\mathcal{L}(\theta) = -H(p^*) - \text{KL}(p^*\|p_\theta) \leq -H(p^*)$ , with equality iff  $p_\theta = p^*$  (by Gibbs' inequality); therefore the expected score is maximised *if and only if* the forecaster reports the true distribution, which is the definition of a strictly proper scoring rule.  $\square$

## C: Parametric Analysis and Small Examples

### C1. Effect of Temperature on NLL and Brier Score

#### Solution

**(a) Large  $T$  (underconfident, uniform distribution):**

- **ECE:** increases – probabilities are now below empirical accuracy (underconfident); reliability diagram lies above the diagonal.
- **NLL:** increases because all class probabilities are pushed toward  $1/C$ , reducing the probability on the true class.
- **Brier:** increases because the predicted vector moves away from the one-hot label.

**(b) Small  $T$  (overconfident, sharp distribution):**



- **ECE:** increases – the model is overconfident; reliability diagram lies below the diagonal.
- **NLL:** quickly increases as occasionally wrong but very confident predictions incur enormous  $-\log p$  penalties; NLL is most sensitive here.
- **Brier:** increases more moderately than NLL because the squared penalty is bounded in  $[0, 1]$ .

**Key insight.** The minimum of all three metrics as a function of  $T$  is near the truly calibrated  $T$ . NLL is sharpest about this minimum (tail-sensitivity).

## C2. Behaviour Under Class Imbalance

### Solution

All predictions fall in the high-confidence bin; mean confidence = 0.95, empirical accuracy in that bin =  $1 - \pi = 0.95$ .

(a) **ECE:**

$$\text{ECE} = \frac{N}{N} |0.95 - 0.95| = \boxed{0}.$$

(b) **Brier score.** Let  $\hat{p}$  be the probability assigned to class 1 (positive). The classifier always outputs  $\hat{p} = 0.05$ .

$$\bar{B} = (1-\pi)(0.05-0)^2 + \pi(0.05-1)^2 = 0.95 \times 0.0025 + 0.05 \times 0.9025 = 0.0024 + 0.0451 = \boxed{0.0475}.$$

(c) **NLL.** Using natural log:

$$\text{NLL} = -(0.95 \ln 0.95 + 0.05 \ln 0.05) \approx -(0.95 \times (-0.0513) + 0.05 \times (-2.996)) = 0.0487 + 0.1498 \approx \boxed{0.199}.$$

(d) The classifier is *technically calibrated* ( $\text{ECE} = 0$ ) because its constant prediction  $\hat{p} = 0.95$  matches the majority class frequency. However, it is *completely useless*: it never identifies a positive case (recall = 0). This illustrates that **calibration alone is insufficient**; *sharpness* (being more confident for positives than negatives) is also required for an informative probabilistic classifier.

## C3. Isotonic Regression vs. Histogram Binning under Small Calibration Sets

### Solution

(a) With  $K = 10$  equal-width bins and  $M = 50$  samples:

$$\mathbb{E}[\text{samples per bin}] = 50/10 = \boxed{5}.$$

(b)

- **Histogram binning:**  $K = 10$  free parameters; with  $\approx 5$  samples per bin, accuracy estimates are high-variance but the method cannot interpolate between bins.
- **Isotonic regression:** uses the PAV algorithm and effectively selects the number of steps from the data. With  $M = 50$  it may produce many steps, potentially overfitting calibration noise.

(c) **Recommendation:** With  $M = 50$ , prefer **temperature scaling** (1 free parameter) over isotonic regression. If a non-parametric method is needed, use equal-frequency histogram binning with  $K \leq 5$  to ensure at least 10 samples per bin. Isotonic regression becomes competitive when  $M \gtrsim 500$ .

#### C4. Discrete Cross-Entropy = Entropy + KL: Numeric Verification

##### Solution

(a) **Entropy of  $p^*$ :**

$$\begin{aligned} H(p^*) &= -(0.50 \ln 0.50 + 0.30 \ln 0.30 + 0.20 \ln 0.20) \\ &= -(0.50 \times (-0.693) + 0.30 \times (-1.204) + 0.20 \times (-1.609)) \\ &= -(-0.347 - 0.361 - 0.322) = \boxed{1.030 \text{ nats}}. \end{aligned}$$

(b) **KL divergence  $\text{KL}(p^* \| p_\theta)$ :**

$$\begin{aligned} \text{KL}(p^* \| p_\theta) &= 0.50 \ln \frac{0.50}{0.70} + 0.30 \ln \frac{0.30}{0.20} + 0.20 \ln \frac{0.20}{0.10} \\ &= 0.50 \times (-0.336) + 0.30 \times (0.405) + 0.20 \times (0.693) \\ &= -0.168 + 0.122 + 0.139 = \boxed{0.093 \text{ nats}}. \end{aligned}$$

(Using  $\ln(0.50/0.70) = \ln 0.50 - \ln 0.70 = -0.693 - (-0.357) = -0.336$ , etc.)

(c) **Cross-entropy  $H(p^*, p_\theta)$  computed directly:**

$$\begin{aligned} H(p^*, p_\theta) &= -(0.50 \ln 0.70 + 0.30 \ln 0.20 + 0.20 \ln 0.10) \\ &= -(0.50 \times (-0.357) + 0.30 \times (-1.609) + 0.20 \times (-2.303)) \\ &= -(-0.179 - 0.483 - 0.461) = \boxed{1.123 \text{ nats}}. \end{aligned}$$

(d) **Verification:**

$$H(p^*) + \text{KL}(p^* \| p_\theta) = 1.030 + 0.093 = \boxed{1.123} = H(p^*, p_\theta). \checkmark$$

(Tiny rounding errors at the third decimal place are acceptable.)

(e)  $H(p^*, p_\theta) = 1.123 > H(p^*) = 1.030$ . The model is *adding* uncertainty (from 1.030 to 1.123 nats) because it assigns probability mass to the wrong classes relative to the truth. The gap = 0.093 nats is the KL divergence measuring the mismatch between model and truth. A perfect model would have  $H(p^*, p_\theta) = H(p^*)$ .

## D: Applied Problems and Notebook Extensions

### D1. Medical Diagnostic Classifier

#### Solution

- (a)  $\text{ECE} = 0.14$  means the model is systematically overconfident: when it reports 90% probability of the condition, only about 76% of patients actually have it. In clinical practice, overconfident predictions drive over-treatment of false positives and under-investigation of patients whose true probability is understated. Patient outcomes make this directly harmful.
- (b)  $T^* = 1.8 > 1$ : dividing logits by 1.8 makes the softmax softer (less peaked). The optimal temperature being  $> 1$  means the *original* model was **overconfident**: its predicted probabilities were too extreme (too close to 0 or 1) relative to the true posterior.
- (c) Temperature scaling applies  $\text{softmax}(\mathbf{z}/T)$ , which preserves the argmax of  $\mathbf{z}$  (positive  $T$  does not change the ordering of logit components). Therefore the predicted class for each sample is identical before and after scaling, leaving accuracy unchanged.
- (d)
- (i) **Decision support:** A probability of 55% vs. 95% changes clinical management dramatically (watchful waiting vs. immediate intervention). A binary label discards this actionable information.
  - (ii) **Uncertainty propagation:** The classifier output may feed into a larger pipeline (cost-benefit analysis, resource allocation). Without calibrated probabilities, uncertainty cannot be properly propagated or communicated.

### D2. Notebook Extension: Comparing All Four Calibration Methods

#### Solution

- (a) Temperature scaling has one free parameter ( $T$ ), whereas isotonic regression has potentially many steps. On a large CIFAR-100 network, miscalibration is approximately *global* (uniformly overconfident), which a single  $T$  corrects very efficiently. Isotonic regression, despite its flexibility, can overfit the calibration set's noise and may not generalise as well to the test set when the miscalibration pattern is simple.
- (b) **Experimental plan.** Split the labelled data into training (60%), calibration (20%), and test (20%); train the base model on training only. For each calibration method (temperature scaling, Platt scaling, histogram binning with  $K = 15$  equal-frequency bins, isotonic regression), fit on the calibration set using the frozen base model's logits. Evaluate ECE, NLL, and Brier score on the held-out test set. All comparisons are on the same frozen logits; the test set is never used for fitting.
- (c) With  $M = 100$ , isotonic regression has very few samples per step and likely overfits. As  $M$  grows to 5000, isotonic regression gains sufficient data to estimate the calibration mapping reliably at all confidence levels, and its greater flexibility allows it to correct non-global miscalibration patterns that temperature scaling cannot. Relative performance is therefore expected to shift in favour of isotonic regression as  $M$  increases.

## 2 Part II – Conformal Prediction

### A: Conceptual Questions

#### A1. What Conformal Prediction Guarantees

##### Solution

(a) Split conformal prediction guarantees:

$$\Pr(Y \in \mathcal{T}(X)) \geq 1 - \alpha,$$

where the probability is over both the calibration set and the new test point  $(X, Y)$  drawn from the same exchangeable distribution.

(b)

- **Marginal coverage:**  $\Pr(Y \in \mathcal{T}(X)) \geq 1 - \alpha$  on average over the joint distribution of  $(X, Y)$ . A single  $\hat{q}$  is applied to all inputs.
- **Conditional coverage:**  $\Pr(Y \in \mathcal{T}(X) \mid X = x) \geq 1 - \alpha$  for every specific  $x$ . Strictly stronger.

**Counterexample.** A binary classifier on easy and hard images (50/50 split). A single  $\hat{q}$  calibrated on the full dataset may over-cover easy images ( $\approx 99\%$ ) and under-cover hard ones ( $\approx 80\%$ ), while the overall average is exactly 90%.

(c) The required assumption is **exchangeability** of calibration and test data (i.i.d. is sufficient but stronger). Exchangeability means the joint distribution of all  $n + 1$  scores is invariant to permutations, without specifying its form. This is weaker than assuming a parametric family (Gaussian, logistic, etc.), because any marginal distribution with a symmetric dependence structure satisfies it.

#### A2. The Role of the Nonconformity Score

##### Solution

(a)

- **Threshold score**  $s_{\text{th}} = 1 - p(y \mid x)$ : the probability mass *not* assigned to the true label. Score  $\approx 0$  when the model is confident in the true label;  $\approx 1$  when the model is wrong or uncertain.
- **APS score:** accumulates all probabilities of classes ranked at least as high as the true label. Measures “how deep in the ranked list” the true label sits. A true label ranked first gives a score equal to  $p(y \mid x)$ ; ranked last gives 1.

A larger score means the model finds the true label *less conforming* (less likely).

(b) Classes ranked:  $A(0.70) > B(0.20) > C(0.10)$ . APS for  $y = B$  sums all classes with  $p \geq p(B) = 0.20$ : that is  $A$  and  $B$ :

$$s_{\text{APS}}(x, B) = 0.70 + 0.20 = 0.90.$$

(c) APS is preferred when class probabilities are well-calibrated. In that regime, APS tends to produce smaller average set sizes on unambiguous inputs (the true label is typically ranked

first, giving a small APS score), while avoiding unnecessary inclusion of low-ranked classes that the threshold method may include if  $\hat{q}$  is large.

### A3. Conformal Prediction vs. Bayesian Credible Intervals

#### Solution

- (a) A Bayesian credible interval requires a *correctly specified* probabilistic model (likelihood and prior). Under model misspecification the stated credibility level need not correspond to any frequentist coverage guarantee. Conformal prediction requires only exchangeability; no parametric assumption on the model or data.
- (b) The conformal guarantee is a **frequentist, finite-sample** statement:  $\Pr(Y \in \mathcal{T}(X)) \geq 1 - \alpha$  holds for any sample size  $n$  and any exchangeable distribution, with no approximation. A Bayesian credible interval expresses a posterior belief under a model; it is not a frequency guarantee over repeated sampling.
- (c) Yes. Conformal prediction can wrap any model, including a Bayesian one. Applied on top of a Bayesian posterior, it adds a *guaranteed frequentist coverage* regardless of whether the prior or likelihood is misspecified. The Bayesian model provides informative scores (improving efficiency), while the conformal layer ensures stated coverage is valid.

### A4. Coverage vs. Efficiency Trade-off

#### Solution

- (a) Returning  $\mathcal{T}(x) = \mathcal{Y}$  trivially satisfies  $Y \in \mathcal{T}(X)$  with probability 1. But the set carries zero information: it tells the user nothing about which label is most likely and is indistinguishable from making no prediction.
- (b) An *efficient* conformal predictor achieves the target coverage  $\geq 1 - \alpha$  while minimising the average set size  $\mathbb{E}[|\mathcal{T}(X)|]$  (classification) or average interval length  $\mathbb{E}[|C(X)|]$  (regression). The ideal is to produce singleton sets (size 1) for every input, which is achievable only with a perfect model.
- (c) Model quality directly controls efficiency. A good model assigns high probability to the true label, producing small nonconformity scores on the calibration set and thus a small  $\hat{q}$ . A random classifier produces approximately uniform scores, forcing a large  $\hat{q}$  to achieve coverage; at the same coverage level it yields average set sizes close to  $|\mathcal{Y}|$ .

### A5. Distribution Shift and Conformal Validity

#### Solution

- (a) The coverage proof uses exchangeability: the test nonconformity score comes from the same distribution as the calibration scores. Under distribution shift, test inputs come from a different distribution; the test scores have a different distribution than calibration scores, the exchangeability argument breaks down, and the guarantee no longer holds.
- (b) Under Gaussian noise, the classifier assigns *lower* probabilities to the true label on noisy test images than on clean calibration images (the model is more uncertain on noisy data).

The threshold  $\hat{q}$  calibrated on clean data is too small for noisy inputs; many test scores exceed  $\hat{q}$ . Therefore empirical coverage will be **below** the nominal level.

(c) Strategies:

- **Recalibrate on in-distribution data:** collect a calibration set from the shifted domain and recompute  $\hat{q}$ .
- **Weighted conformal prediction:** reweight calibration scores by the likelihood ratio  $p_{\text{test}}(x)/p_{\text{cal}}(x)$  before computing the quantile (covariate-shift correction).
- **Mondrian conformal prediction:** stratify by difficulty/domain features, computing a separate  $\hat{q}$  per stratum.

## A6. Designing a Calibration Dataset

### Solution

(a) **Four key criteria:**

- (i) **Exchangeability with the test data.** Calibration and test points must come from the same distribution (or a distribution symmetric with respect to permutations). This is the fundamental mathematical requirement for the coverage proof.
- (ii) **No data leakage.** The calibration set must be *disjoint* from the training set. If any training sample appears in calibration, the model has already “seen” it; its nonconformity scores will be artificially small, causing the calibrated  $\hat{q}$  to be too low and the coverage guarantee to fail.
- (iii) **Temporal or group structure.** In time-series or grouped data, naive random splitting violates exchangeability. Use a *temporal hold-out* (calibration data come after training data in time) or a *group-stratified split* (no group appears in both training and calibration) to preserve the assumption.
- (iv) **Sufficient size.** The finite-sample overcoverage is  $1/(n + 1)$ . To limit overcoverage to  $\leq 0.005$ , at least  $n \geq 199$  calibration samples are required. Smaller calibration sets also produce noisier  $\hat{q}$  estimates, especially for Mondrian methods that require per-class calibration.

(b) If calibration data are not exchangeable with test data, the conformal quantile  $\hat{q}$  may be too small or too large for the test distribution.

**Engineering example.** Suppose a structural load model is calibrated on components from a *summer* production run and tested on *winter* components. If winter materials have higher variability, test nonconformity scores will exceed  $\hat{q}$ , leading to *under-coverage*: fewer than  $1 - \alpha$  of intervals contain the true load. The coverage guarantee, which was proved under exchangeability, no longer holds.

**Medical example.** A classifier calibrated on data from a single hospital may not be exchangeable with data from a different hospital (different demographics, imaging protocols, etc.). Coverage may fail in either direction.

(c) The choice of  $\alpha$  should reflect the *cost of a coverage error*:

- $\alpha = 0.10$  (90% coverage) is typical for exploratory prediction sets where missing the true label occasionally is acceptable, e.g., an initial screening triage.
- $\alpha = 0.10$  would be too lax for safety-critical decisions such as drug dosing or bridge load certification, where  $\alpha = 0.01$  or  $\alpha = 0.001$  may be more appropriate.

$1 - \alpha$  means: “over many test inputs drawn from the same distribution, the true label will be in the prediction set at least a fraction  $1 - \alpha$  of the time.”

(d) Using the same data for both calibration and testing violates the independence required by the coverage proof. The model has effectively been evaluated on the data used to set  $\hat{q}$ , causing  $\hat{q}$  to be optimistically small (since the calibration scores match the test scores exactly). Empirical coverage will appear correct in-sample but the guarantee does not transfer to genuinely unseen data.

## B: Mathematical Derivations

### B1. The Finite-Sample Coverage Proof

#### Solution

(a) Under exchangeability, all  $n + 1$  scores  $\{s_1, \dots, s_{n+1}\}$  are jointly exchangeable. Every permutation is equally likely, so the rank of  $s_{n+1}$  in the sorted sequence  $s_{(1)} \leq \dots \leq s_{(n+1)}$  is

$$\text{rank}(s_{n+1}) \sim \text{Uniform}\{1, 2, \dots, n + 1\}.$$

(b) Let  $k = \lceil (1 - \alpha)(n + 1) \rceil$  and  $\hat{q} = s_{(k)}$ .

$$\begin{aligned} \Pr(s_{n+1} > \hat{q}) &= \Pr(\text{rank}(s_{n+1}) > k) \\ &= \frac{n + 1 - k}{n + 1}. \end{aligned}$$

Since  $k = \lceil (1 - \alpha)(n + 1) \rceil \geq (1 - \alpha)(n + 1)$ , we have  $n + 1 - k \leq n + 1 - (1 - \alpha)(n + 1) = \alpha(n + 1)$ , so:

$$\Pr(s_{n+1} > \hat{q}) \leq \frac{\alpha(n + 1)}{n + 1} = \alpha.$$

Equivalently,  $\Pr(s_{n+1} \leq \hat{q}) \geq 1 - \alpha$ .

(c) The exact coverage is:

$$\Pr(s_{n+1} \leq \hat{q}) = \frac{k}{n + 1} = \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n + 1} \leq 1 - \alpha + \frac{1}{n + 1}.$$

$$\text{Coverage} \in \left[ 1 - \alpha, 1 - \alpha + \frac{1}{n + 1} \right].$$

This shows conformal prediction is *slightly conservative*: it can over-cover by up to  $1/(n + 1)$ . For large  $n$  this is negligible; for small  $n$ , prediction sets may be slightly larger than necessary.

## B2. Split Conformal Regression: Interval Construction

### Solution

(a) A new  $y$  is included in the prediction set iff  $s(x, y) \leq \hat{q}$ :

$$y \in C(x) \iff |y - \hat{y}(x)| \leq \hat{q} \iff -\hat{q} \leq y - \hat{y}(x) \leq \hat{q} \iff \hat{y}(x) - \hat{q} \leq y \leq \hat{y}(x) + \hat{q}.$$

This is exactly the interval  $[\hat{y}(x) - \hat{q}, \hat{y}(x) + \hat{q}]$ .

(b) With  $n = 99$  and  $\alpha = 0.10$ :

$$k = \lceil (1 - 0.10)(99 + 1) \rceil = \lceil 0.90 \times 100 \rceil = \lceil 90 \rceil = 90.$$

$$\hat{q} = s_{(90)} \text{ (the 90th smallest of the 99 calibration residuals).}$$

(c) Constant-width intervals are problematic when residual variance depends on  $x$  (**heteroscedasticity**). For example, in house-price prediction, expensive houses have larger residuals than cheap ones; a single  $\hat{q}$  over-covers cheap houses and under-covers expensive ones (conditional coverage fails). **Conformalized Quantile Regression (CQR)** addresses this by training a model to predict conditional quantiles, yielding adaptive-width intervals.

## B3. Effect of Calibration Set Size on the Quantile

### Solution

(a) By the Glivenko-Cantelli theorem, the empirical CDF converges uniformly to  $F$ . Therefore:

$$\hat{q} \xrightarrow{a.s.} F^{-1}(1 - \alpha) \text{ as } n \rightarrow \infty.$$

(b) The conformal quantile uses order statistic  $k = \lceil (1 - \alpha)(n + 1) \rceil$ , which corresponds to the empirical quantile level  $k/n \approx (1 - \alpha)(1 + 1/n) > 1 - \alpha$ . This introduces a small **upward bias**:

$$\mathbb{E}[\hat{q}] \approx F^{-1}\left(\frac{k}{n+1}\right) \geq F^{-1}(1 - \alpha).$$

The upward bias is what makes conformal prediction conservative: the estimated threshold is at least as large as the true quantile.

(c) Requiring  $1/(n + 1) \leq 0.005$ :

$$n + 1 \geq 200 \implies n \geq 199.$$

A calibration set of at least 199 points limits finite-sample overcoverage to below 0.5 percentage points.



## B4. APS Score: Computation and Coverage

### Solution

(a) Rank classes by descending probability:  $A(0.50) > B(0.30) > C(0.15) > D(0.05)$ . APS for label  $y$  accumulates probabilities of all classes ranked at least as high:

$$s(x, A) = 0.50$$

$$s(x, B) = 0.50 + 0.30 = 0.80$$

$$s(x, C) = 0.50 + 0.30 + 0.15 = 0.95$$

$$s(x, D) = 0.50 + 0.30 + 0.15 + 0.05 = 1.00$$

(b) Include  $y$  if  $s(x, y) \leq 0.85$ :

$$\mathcal{T}(x) = \{A, B\}$$

since  $s(x, A) = 0.50 \leq 0.85$  and  $s(x, B) = 0.80 \leq 0.85$ , but  $s(x, C) = 0.95 > 0.85$ .

(c) 90% coverage means  $\Pr(Y \in \mathcal{T}(X)) \geq 0.90$ . Out of 1000 test inputs, at least 900 true labels are expected to be in the prediction set. The remaining  $\leq 100$  are coverage errors. This is a marginal statement; individual inputs may have higher or lower conditional coverage.

## C: Parametric Analysis

### C1. Effect of $\alpha$ on Set Size and Coverage

#### Solution

(a) As  $\alpha$  increases, the desired coverage  $1 - \alpha$  decreases. This means  $\hat{q}$  is computed at a lower quantile of calibration scores (a smaller  $k$  in the rank formula). A smaller  $\hat{q}$  means fewer classes satisfy  $s(x, y) \leq \hat{q}$ , so the **average set size decreases** with  $\alpha$ . Intuitively, lower coverage requirements allow the predictor to be more selective.

(b) The finite-sample guarantee  $\Pr(Y \in \mathcal{T}(X)) \geq 1 - \alpha$  holds simultaneously for all  $\alpha$  under exchangeability. Both methods therefore track the nominal level from above for all tested  $\alpha$ .

(c) The Digits classifier achieves  $\approx 97\%$  accuracy. Calibration threshold scores  $1 - p(y | x)$  are typically very small ( $\ll 0.05$ ) because the model is almost always correct with high confidence; even a very small  $\hat{q}$  yields singleton sets. However, at larger  $\alpha$ ,  $\hat{q}$  drops so low that in the discrete setting a small fraction of test scores may exceed it, causing marginal under-coverage. APS scores are larger for the same correct predictions (they equal  $p(y | x)$  rather than  $1 - p(y | x)$ ), making their threshold more conservative and under-coverage less likely.

## C2. Mondrian Conformal Prediction

### Solution

(a) Mondrian conformal provides **class-conditional coverage**:

$$\Pr(Y \in \mathcal{T}(X) \mid Y = c) \geq 1 - \alpha \quad \forall c \in \mathcal{Y}.$$

This is strictly stronger than marginal coverage: coverage holds within each class separately, not just on average over all classes.

(b) Overcoverage for class  $c$  is  $\leq 1/(n_c + 1)$ :

- Class 1 ( $n_1 = 200$ ): overcoverage  $\leq 1/201 \approx 0.005$  – reliable.
- Class 2 ( $n_2 = 20$ ): overcoverage  $\leq 1/21 \approx 0.048$  – unreliable;  $\hat{q}_2$  is noisy and effective coverage could substantially exceed  $1 - \alpha$ .

To limit overcoverage to 0.005:  $n_c + 1 \geq 200$ , i.e.,  $\boxed{n_c \geq 199}$  per class.

(c) **Medical diagnosis with rare diseases.** A classifier trained on 9 common conditions and 1 rare one will have a single  $\hat{q}$  dominated by common-class scores. The rare class, with harder and more variable inputs, will be systematically under-covered. Mondrian conformal, with a per-class  $\hat{q}_c$ , ensures valid coverage even for the rare class (given enough calibration samples).

## C3. CQR vs. Standard Conformal

### Solution

(a) The CQR score  $s(x, y) = \max(\hat{q}_\alpha^-(x) - y, y - \hat{q}_\alpha^+(x))$  is *positive* when  $y$  lies outside the raw quantile interval  $[\hat{q}_\alpha^-(x), \hat{q}_\alpha^+(x)]$ , and the magnitude measures how far outside. A negative value (meaning  $y$  is inside) is clipped to a large negative number; the max effectively measures the signed distance to the nearest quantile boundary.

(b) The prediction set includes all  $y$  with  $s(x, y) \leq \hat{q}$ :

$$\begin{aligned} s(x, y) \leq \hat{q} &\iff \hat{q}_\alpha^-(x) - y \leq \hat{q} \text{ and } y - \hat{q}_\alpha^+(x) \leq \hat{q} \\ &\iff \hat{q}_\alpha^-(x) - \hat{q} \leq y \leq \hat{q}_\alpha^+(x) + \hat{q}. \end{aligned}$$

$$\boxed{C(x) = [\hat{q}_\alpha^-(x) - \hat{q}, \hat{q}_\alpha^+(x) + \hat{q}]},$$

with width  $(\hat{q}_\alpha^+(x) - \hat{q}_\alpha^-(x)) + 2\hat{q}$  – *adaptive* because the first term varies with  $x$ .

(c) Wider CQR intervals in the high- $x$  region indicate that the conditional variance of  $Y$  increases with  $x$  (the data are **heteroscedastic**). The quantile model captures this local uncertainty; CQR translates it into wider sets. Standard conformal regression would apply a constant  $\hat{q}$ , over-covering the low- $x$  region and under-covering the high- $x$  region (conditional coverage failure).

## D: Applied Problems and Notebook Extensions

### D1. Structural Load Prediction

#### Solution

(a) “For any new structural component with features  $x$ , we are *at least 95% confident* that its true maximum load lies within  $\hat{y}(x) \pm 8.3$  kN. Over many such components, at least 95 out of every 100 true loads will be captured. This is a finite-sample, distribution-free guarantee, not a model-dependent approximation.”

(b) Recommend **Conformalized Quantile Regression (CQR)**. Instead of a residual-based constant  $\hat{q}$ , one trains quantile models  $[\hat{q}_{0.025}(x), \hat{q}_{0.975}(x)]$  and conformally adjusts them. The result is an *adaptive-width* interval: wider for variable components, narrower for predictable ones, while maintaining the  $\geq 95\%$  coverage guarantee.

(c) With a new alloy the residual distribution may differ from the calibration data. If the new alloy has higher variability (higher residuals),  $\hat{q} = 8.3$  kN will be too small and the interval will **under-cover** (fewer than 95% of components captured). The exchangeability assumption is violated and the guarantee no longer holds.

(d) From B3: overcoverage  $\leq 1/(n+1) \leq 0.005$  requires  $n+1 \geq 200$ :

$$n \geq 199 \text{ new components must be tested for recalibration.}$$

### D2. Notebook Analysis: Classification Results

#### Solution

(a) Yes, both curves above nominal is *expected* from the finite-sample guarantee: coverage  $\geq 1 - \alpha + 1/(n+1)$ . Empirical coverage on a finite test set can dip slightly below nominal due to *random test-set variability*—this is a fluctuation, not a violation of the guarantee. Averaging over many test repetitions, both curves would lie above nominal.

(b) The Digits classifier is highly accurate ( $\approx 97\%$ ), so calibration scores are concentrated near 0. Threshold score  $1 - p(y | x)$  is typically tiny ( $\ll 0.05$ ), allowing a very small  $\hat{q}$  and frequent singleton sets – appearing more efficient. APS scores equal  $p(y | x)$  for a top-ranked true label, which is larger; this leads to slightly bigger  $\hat{q}$  and larger sets. On harder datasets with ambiguous classes, APS would outperform Threshold by avoiding unnecessary inclusion of loosely ranked classes.

(c) Mondrian calibration per class:

```
import numpy as np

def mondrian_calibrate(cal_probs, cal_labels, alpha, n_classes):
    q_hat = {}
    for c in range(n_classes):
        mask = (cal_labels == c)
        scores_c = 1 - cal_probs[mask, c] # threshold score
        n_c = mask.sum()
        k = int(np.ceil((1 - alpha) * (n_c + 1)))
        k = min(k, n_c)
        q_hat[c] = np.sort(scores_c)[k - 1]
```

```

    return q_hat

def mondrian_predict(test_probs, q_hat, n_classes):
    sets = []
    for p in test_probs:
        pred_set = [c for c in range(n_classes)
                     if 1 - p[c] <= q_hat[c]]
        sets.append(pred_set)
    return sets

```

(d) With a random classifier all probabilities  $= 1/C = 0.1$ , so all calibration scores  $s_i = 1 - 0.1 = 0.9$ . The calibration quantile  $\hat{q} \approx 0.9$  regardless of  $\alpha$ . Every class satisfies  $1 - 0.1 = 0.9 \leq 0.9$ , so all 10 classes are included in every prediction set. The efficiency plot would be a flat line at 10. Coverage would still hold (the distribution-free guarantee remains valid), but the sets are completely uninformative.

### D3. Regression and Distribution Shift

#### Solution

(a)

- (i) **Finite test-set variability.** The reported coverage is the empirical fraction on a finite test set; random fluctuations around the true probability can push the estimate slightly below  $1 - \alpha$ , even when the true probability satisfies the guarantee.
- (ii) **Small calibration set relative to test set.** The conformal guarantee  $\Pr(Y \in C(X)) \geq 1 - \alpha$  is a *marginal* statement over the joint randomness of calibration and test data. For any *fixed* calibration set, the conditional coverage may differ from  $1 - \alpha$ . With a small calibration set (e.g.,  $n \lesssim 100$ ),  $\hat{q}$  is noisy: one unlucky calibration split can produce a  $\hat{q}$  that is too small for the particular test set, yielding empirical coverage below nominal. Note that both standard conformal and CQR share the same distribution-free guarantee; CQR's quantile model quality affects *interval width* (efficiency), not the validity of the coverage guarantee itself.

(b) **Experimental plan.** Generate the shift by adding i.i.d. Gaussian noise  $\mathcal{N}(0, \sigma^2)$  to each pixel of the Digits test images for  $\sigma \in \{0, 5, 15, 30\}$  (on a 0–16 scale). Report empirical coverage  $\hat{c} = N^{-1} \sum_i \mathbf{1}(y_i \in \mathcal{T}(x_i))$  and average set size for each  $\sigma$ . Compute a 95% Wilson confidence interval for the coverage proportion; if the lower bound falls below  $1 - \alpha$ , the guarantee has statistically broken. Plot coverage vs.  $\sigma$  to visualise the degradation curve.

(c) Define difficulty buckets by top-1 probability: bucket 0:  $\hat{p}_{\max} > 0.9$ ; bucket 1:  $0.7 < \hat{p}_{\max} \leq 0.9$ ; bucket 2:  $\hat{p}_{\max} \leq 0.7$ .

```

def bucket_id(prob_vec, thresholds=[0.9, 0.7]):
    p_max = prob_vec.max()
    if p_max > thresholds[0]: return 0
    elif p_max > thresholds[1]: return 1
    else: return 2

```

```

# Calibration
q_hat_bucket = {}
for b in range(3):
    mask      = [bucket_id(p) == b for p in cal_probs]
    scores_b = [1 - cal_probs[i, cal_labels[i]]
                for i in range(len(cal_labels)) if mask[i]]
    n_b = len(scores_b)
    k   = int(np.ceil((1 - alpha) * (n_b + 1)))
    q_hat_bucket[b] = np.sort(scores_b)[min(k, n_b) - 1]

# Prediction
def bucketed_predict(test_prob, q_hat_bucket, n_classes):
    b = bucket_id(test_prob)
    return [c for c in range(n_classes)
            if 1 - test_prob[c] <= q_hat_bucket[b]]

```

## E: Analytical Table Problems

### E1. Fill-the-Table: Classification Conformal Prediction

#### Solution

(i) **Threshold scores**  $s_{\text{th},i} = 1 - p(y_i | x_i)$ :

| $i$ | $p(A)$ | $p(B)$ | $p(C)$ | $y_i$ | $s_{\text{th},i}$          |
|-----|--------|--------|--------|-------|----------------------------|
| 1   | 0.70   | 0.20   | 0.10   | $A$   | $1 - 0.70 = \mathbf{0.30}$ |
| 2   | 0.50   | 0.30   | 0.20   | $B$   | $1 - 0.30 = \mathbf{0.70}$ |
| 3   | 0.15   | 0.65   | 0.20   | $B$   | $1 - 0.65 = \mathbf{0.35}$ |
| 4   | 0.10   | 0.20   | 0.70   | $C$   | $1 - 0.70 = \mathbf{0.30}$ |
| 5   | 0.40   | 0.35   | 0.25   | $A$   | $1 - 0.40 = \mathbf{0.60}$ |

(ii) **APS scores**  $s_{\text{APS},i} = \sum_{y': p(y'|x_i) \geq p(y_i|x_i)} p(y' | x_i)$ :

| $i$ | $y_i$ | Classes summed ( $p \geq p(y_i)$ )                               | $s_{\text{APS},i}$            |
|-----|-------|------------------------------------------------------------------|-------------------------------|
| 1   | $A$   | Rank: $A(0.70) > B(0.20) > C(0.10)$ . $p \geq 0.70$ : $\{A\}$    | <b>0.70</b>                   |
| 2   | $B$   | Rank: $A(0.50) > B(0.30) > C(0.20)$ . $p \geq 0.30$ : $\{A, B\}$ | $0.50 + 0.30 = \mathbf{0.80}$ |
| 3   | $B$   | Rank: $B(0.65) > C(0.20) > A(0.15)$ . $p \geq 0.65$ : $\{B\}$    | <b>0.65</b>                   |
| 4   | $C$   | Rank: $C(0.70) > B(0.20) > A(0.10)$ . $p \geq 0.70$ : $\{C\}$    | <b>0.70</b>                   |
| 5   | $A$   | Rank: $A(0.40) > B(0.35) > C(0.25)$ . $p \geq 0.40$ : $\{A\}$    | <b>0.40</b>                   |

(iii) **Quantile index:**

$$k = \lceil (n+1)(1-\alpha) \rceil = \lceil 6 \times 0.80 \rceil = \lceil 4.8 \rceil = \boxed{5}.$$

(iv) **Compute  $\hat{q}$  for each method.**

Threshold scores sorted: 0.30, 0.30, 0.35, 0.60, 0.70.

$$\boxed{\hat{q}_{\text{th}} = s_{(5)} = 0.70}.$$

APS scores sorted: 0.40, 0.65, 0.70, 0.70, 0.80.

$$\hat{q}_{\text{APS}} = s_{(5)} = 0.80.$$

**(v) Test Point 1:**  $\hat{\mathbf{p}} = (0.55, 0.30, 0.15)$ .

*Threshold* (include  $y$  if  $p(y) \geq 1 - \hat{q}_{\text{th}} = 0.30$ ):

- $p(A) = 0.55 \geq 0.30$ : ✓
- $p(B) = 0.30 \geq 0.30$ : ✓ (boundary; use  $\geq$ )
- $p(C) = 0.15 < 0.30$ : ✗

$\mathcal{T}_{\text{th}} = \{A, B\}$ .

*APS* (include  $y$  if  $s_{\text{APS}} \leq 0.80$ ). Rank:  $A(0.55) > B(0.30) > C(0.15)$ .

- $s_{\text{APS}}(A) = 0.55 \leq 0.80$ : ✓
- $s_{\text{APS}}(B) = 0.55 + 0.30 = 0.85 > 0.80$ : ✗
- $s_{\text{APS}}(C) = 0.55 + 0.30 + 0.15 = 1.00 > 0.80$ : ✗

$\mathcal{T}_{\text{APS}} = \{A\}$ .

**(vi) Test Point 2:**  $\hat{\mathbf{p}} = (0.28, 0.27, 0.45)$ .

*Threshold* (include  $y$  if  $p(y) \geq 0.30$ ):

- $p(A) = 0.28 < 0.30$ : ✗
- $p(B) = 0.27 < 0.30$ : ✗
- $p(C) = 0.45 \geq 0.30$ : ✓

$\mathcal{T}_{\text{th}} = \{C\}$ .

*APS* (include  $y$  if  $s_{\text{APS}} \leq 0.80$ ). Rank:  $C(0.45) > A(0.28) > B(0.27)$ .

- $s_{\text{APS}}(C) = 0.45 \leq 0.80$ : ✓
- $s_{\text{APS}}(A) = 0.45 + 0.28 = 0.73 \leq 0.80$ : ✓
- $s_{\text{APS}}(B) = 0.45 + 0.28 + 0.27 = 1.00 > 0.80$ : ✗

$\mathcal{T}_{\text{APS}} = \{C, A\}$ .

**(vii) Comparison table:**

|        | Threshold set       | APS set             |
|--------|---------------------|---------------------|
| Test 1 | $\{A, B\}$ (size 2) | $\{A\}$ (size 1)    |
| Test 2 | $\{C\}$ (size 1)    | $\{A, C\}$ (size 2) |

**One-sentence comparison.** Threshold includes any class above a fixed probability cutoff (favouring recall), while APS accumulates probability mass along the ranked list (favouring precision); on Test 1, where the runner-up class  $B$  is just at the cutoff, Threshold includes it but APS's cumulative sum already exceeds  $\hat{q}_{\text{APS}}$ , giving a smaller set; on Test 2, the opposite occurs because  $C$ 's mass alone is insufficient to exclude  $A$  under APS.

## E2. Fill-the-Table: Regression Conformal Prediction

### Solution

(i) **Residual scores**  $s_i = |y_i - \hat{y}(x_i)|$ :

| $i$ | $x_i$ | $\hat{y}(x_i)$ | $y_i$ | $s_i =  y_i - \hat{y} $      |
|-----|-------|----------------|-------|------------------------------|
| 1   | 1.0   | 2.5            | 2.8   | $ 2.8 - 2.5  = \mathbf{0.3}$ |
| 2   | 2.0   | 4.0            | 3.3   | $ 3.3 - 4.0  = \mathbf{0.7}$ |
| 3   | 3.0   | 5.5            | 6.1   | $ 6.1 - 5.5  = \mathbf{0.6}$ |
| 4   | 4.0   | 7.0            | 5.8   | $ 5.8 - 7.0  = \mathbf{1.2}$ |
| 5   | 5.0   | 8.5            | 9.8   | $ 9.8 - 8.5  = \mathbf{1.3}$ |

(ii) **Quantile index** with  $\alpha = 0.20$ :

$$k = \lceil (5 + 1)(1 - 0.20) \rceil = \lceil 6 \times 0.80 \rceil = \lceil 4.8 \rceil = 5.$$

Scores sorted in ascending order: 0.3, 0.6, 0.7, 1.2, 1.3.

$$\hat{q} = s_{(5)} = 1.3.$$

(iii) **Prediction intervals**  $C(x) = [\hat{y}(x) - \hat{q}, \hat{y}(x) + \hat{q}]$ :

| Test point | $\hat{y}(x)$ | $C(x)$                                    |
|------------|--------------|-------------------------------------------|
| $x = 2.5$  | 4.75         | $[4.75 - 1.3, 4.75 + 1.3] = [3.45, 6.05]$ |
| $x = 4.5$  | 7.75         | $[7.75 - 1.3, 7.75 + 1.3] = [6.45, 9.05]$ |

(iv) **Edge case** with  $\alpha = 0.10$ :

$$k = \lceil (5 + 1)(1 - 0.10) \rceil = \lceil 6 \times 0.90 \rceil = \lceil 5.4 \rceil = 6.$$

We only have  $n = 5$  calibration scores; the required 6th order statistic does not exist. By the standard convention,  $\hat{q} = +\infty$  in this case, and the prediction interval becomes  $(-\infty, +\infty)$ : every real value is included. **In practice, this means 5 calibration points are insufficient to certify 90% coverage with the  $(n + 1)$  correction.** A minimum of  $n \geq 9$  calibration points is required to achieve a non-trivial result at  $\alpha = 0.10$  (since  $\lceil 10 \times 0.90 \rceil = 9 \leq 9$ ).

(v) **Width interpretation.** Both intervals have the same width  $2\hat{q} = 2.6$ , regardless of whether the input is at  $x = 2.5$  or  $x = 4.5$ . This is a **limitation** of standard conformal regression: the constant width ignores heteroscedasticity (the residuals in our table clearly increase with  $x$ , suggesting higher variance at larger  $x$ ). **Conformalized Quantile Regression (CQR)** produces adaptive-width intervals by modelling conditional quantiles.

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